

PHYSICS 550
Homework #5
Due on Oct 27, 2023

1. G3.13 (10 points)
2. G 3.23 (10 points)
3. G 3.24 (10 points)
4. G 3.37 (10 points)
5. (10 points) A particle of mass M experiences a potential where $V(x) = \infty$ for $x < 0$, $V(x) = -V_0$ for $0 < x < a$ and $V(x) = 0$ for $a < x$.
 - (a) Find the transcendental equation that will give the bound state energies.
 - (b) There are combinations of V_0 and a that *do not* give any bound states. Determine the condition on V_0 and a that means there will be at least one bound state.